

ANAVI I2C TPM 2.0

DATASHEET

Overview

ANAVI I2C TPM 2.0 is an open-source hardware add-on that provides a **Trusted Platform Module (TPM) 2.0** functionality to Raspberry Pi single-board computers via the **I2C** interface. It is designed to deliver hardware-based security, including secure key storage, cryptographic operations, and platform integrity.

The board is built around the **Infineon OPTIGA™ TPM SLB 9673**, a certified TPM 2.0 security controller compliant with modern standards and cryptographic requirements.

The **OPTIGA™ TPM SLB 9673** is a high-end, standardized TPM 2.0 security controller designed for PCs, embedded systems, and IoT devices. It provides strong cryptographic capabilities and future-proof security features including post-quantum protection.

Key Features

ANAVI I2C TPM 2.0

- Open-source hardware design
 - Compatible with Raspberry Pi single board computers via GPIO (1x5 header)
 - Available in:
 - Horizontal version ANAVI-I2C-TPM2-H
 - Vertical (90°) version ANAVI-I2C-TPM2-V
 - Supports SPI communication interface
 - Functions as:
 - Secure key storage
 - Cryptographic processor
 - True Hardware Random Number Generator (TRNG)
 - Made in Plovdiv, Bulgaria, EU
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ANAVI I2C TPM 2.0 Variants

Variant	Mounting	Part Number
ANAVI I2C TPM 2.0 Horizontal	Horizontal	ANAVI-I2C-TPM2-H
ANAVI I2C TPM 2.0 Vertical	Vertical	ANAVI-I2C-TPM2-V

Security Features & Compliance

Infineon OPTIGA TPM SLB 9673 used in ANAVI I2C TPM 2.0 is:

- TCG TPM 2.0 Library Specification Rev. 1.59 and PC Client Platform TPM Profile (PTP) version 1.05
- High confidence level based on Common Criteria and FIPS 140-2 Level 2
- I2C interface (using clock stretching)
- Random Number Generator (RNG)
- Provisioned with 4 Endorsement Keys (EK) and EK certificates
- 24 PCRs (SHA-1, SHA-256 or SHA384)
- RSA, ECC, SHA-1, SHA-256, SHA-384, AES-128, AES-192, AES-256
- Firmware update capability with PQC protection

Electrical & Interface Characteristics

Parameter	Value
Interface	I2C
Supply Voltage	3.3 V
Raspberry Pi Pins	5 pins: 1, 3, 5, 7, 9
PCB Dimensions	16.5 x 12.8mm (0.65 x 0.51 inches)
Temperature Range	-20°C to +85°C

Software & Compatibility

- Native support in Linux distributions, including Raspberry Pi OS
- Works with standard TPM drivers and tools
- Compatible with existing TPM device tree overlay `tpm-slb9673` in the Linux kernel for Raspberry Pi OS
- Supported by development tools and example code

Applications

- IoT and edge devices
 - Secure Raspberry Pi projects
 - Industrial automation
 - Networking equipment (routers, gateways)
 - Servers and embedded systems
 - Smart buildings and home automation
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Mechanical & Integration

- Form factor: compact add-on board
- Connector: 1x5 pin female header (2.54 mm)
- Mounting options: horizontal or vertical
- Compatible with all Raspberry Pi single board computers